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compact1, compact2, compact3

java.util

# Class Locale.Builder

java.lang.Object
    java.util.Locale.Builder

**Enclosing class:**

Locale

---

```
public static final class Locale.Builder
extends Object
```

`Builder` is used to build instances of `Locale` from values configured by the setters. Unlike the `Locale` constructors, the `Builder` checks if a value configured by a setter satisfies the syntax requirements defined by the `Locale` class. A `Locale` object created by a `Builder` is well-formed and can be transformed to a well-formed IETF BCP 47 language tag without losing information.

**Note:** The `Locale` class does not provide any syntactic restrictions on variant, while BCP 47 requires each variant subtag to be 5 to 8 alphanumerics or a single numeric followed by 3 alphanumerics. The method `setVariant` throws `IllformedLocaleException` for a variant that does not satisfy this restriction. If it is necessary to support such a variant, use a Locale constructor. However, keep in mind that a `Locale` object created this way might lose the variant information when transformed to a BCP 47 language tag.

The following example shows how to create a `Locale` object with the `Builder`.

```
Locale aLocale = new Builder().setLanguage("sr").setScript("Latn").setReg
```

Builders can be reused; `clear()` resets all fields to their default values.

**Since:**

1.7

**See Also:**

`Locale.forLanguageTag(java.lang.String)`

## *Constructor Summary*

**Constructors**

| Constructor | Description |
| --- | --- |
| `Builder()` | Constructs an empty Builder. |

## *Method Summary*

**Concrete Methods**

| Modifier and Type | Method | Description |
| --- | --- | --- |
| `Locale.Builder` | `addUnicodeLocaleAttribute(String attribute)` | Adds a unicode locale attribute, if not already present, otherwise has no effect. |
| `Locale` | `build()` | Returns an instance of `Locale` created from the fields set on this builder. |
| `Locale.Builder` | `clear()` | Resets the builder to its initial, empty state. |
| `Locale.Builder` | `clearExtensions()` | Resets the extensions to their initial, |

| | | |
|---|---|---|
| | | empty state. |
| **Locale.Builder** | **removeUnicodeLocaleAttribute**(**String** attribute) | Removes a unicode locale attribute, if present, otherwise has no effect. |
| **Locale.Builder** | **setExtension**(char key, **String** value) | Sets the extension for the given key. |
| **Locale.Builder** | **setLanguage**(**String** language) | Sets the language. |
| **Locale.Builder** | **setLanguageTag**(**String** languageTag) | Resets the Builder to match the provided IETF BCP 47 language tag. |
| **Locale.Builder** | **setLocale**(**Locale** locale) | Resets the `Builder` to match the provided `locale`. |
| **Locale.Builder** | **setRegion**(**String** region) | Sets the region. |
| **Locale.Builder** | **setScript**(**String** script) | Sets the script. |

| | | |
|---|---|---|
| Locale.Builder | setUnicodeLocaleKeyword(String key, String type) | Sets the Unicode locale keyword type for the given key. |
| Locale.Builder | setVariant(String variant) | Sets the variant. |

### Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

## *Constructor Detail*

### Builder

```
public Builder()
```

Constructs an empty Builder. The default value of all fields, extensions, and private use information is the empty string.

## *Method Detail*

### setLocale

```
public Locale.Builder setLocale(Locale locale)
```

Resets the `Builder` to match the provided `locale`. Existing state is discarded.

All fields of the locale must be well-formed, see `Locale`.

Locales with any ill-formed fields cause `IllformedLocaleException` to be thrown, except for the following three cases which are accepted for compatibility reasons:

- Locale("ja", "JP", "JP") is treated as "ja-JP-u-ca-japanese"
- Locale("th", "TH", "TH") is treated as "th-TH-u-nu-thai"
- Locale("no", "NO", "NY") is treated as "nn-NO"

**Parameters:**

locale - the locale

**Returns:**

This builder.

**Throws:**

IllformedLocaleException - if locale has any ill-formed fields.

NullPointerException - if locale is null.

### setLanguageTag

public Locale.Builder setLanguageTag(String languageTag)

Resets the Builder to match the provided IETF BCP 47 language tag. Discards the existing state. Null and the empty string cause the builder to be reset, like clear(). Grandfathered tags (see Locale.forLanguageTag(java.lang.String)) are converted to their canonical form before being processed. Otherwise, the language tag must be well-formed (see Locale) or an exception is thrown (unlike Locale.forLanguageTag, which just discards ill-formed and following portions of the tag).

**Parameters:**

languageTag - the language tag

**Returns:**

This builder.

**Throws:**

IllformedLocaleException - if languageTag is ill-formed

**See Also:**

Locale.forLanguageTag(String)

### setLanguage

public Locale.Builder setLanguage(String language)

Sets the language. If language is the empty string or null, the language in this Builder is removed. Otherwise, the language must be well-formed or an exception is thrown.

The typical language value is a two or three-letter language code as defined in ISO639.

**Parameters:**

language - the language

**Returns:**

This builder.

**Throws:**

IllformedLocaleException - if language is ill-formed

### setScript

public Locale.Builder setScript(String script)

Sets the script. If script is null or the empty string, the script in this Builder is removed. Otherwise, the script must be well-formed or an exception is thrown.

The typical script value is a four-letter script code as defined by ISO 15924.

**Parameters:**

script - the script

**Returns:**

This builder.

**Throws:**

IllformedLocaleException - if script is ill-formed

### setRegion

public Locale.Builder setRegion(String region)

Sets the region. If region is null or the empty string, the region in this Builder is removed. Otherwise, the region must be well-formed or an exception is thrown.

The typical region value is a two-letter ISO 3166 code or a three-digit UN M. 49 area code.

The country value in the Locale created by the Builder is always normalized to upper case.

**Parameters:**

region - the region

**Returns:**

This builder.

**Throws:**

IllformedLocaleException - if region is ill-formed

### setVariant

```
public Locale.Builder setVariant(String variant)
```

Sets the variant. If variant is null or the empty string, the variant in this `Builder` is removed. Otherwise, it must consist of one or more well-formed subtags, or an exception is thrown.

**Note:** This method checks if `variant` satisfies the IETF BCP 47 variant subtag's syntax requirements, and normalizes the value to lowercase letters. However, the `Locale` class does not impose any syntactic restriction on variant, and the variant value in `Locale` is case sensitive. To set such a variant, use a Locale constructor.

**Parameters:**

`variant - the variant`

**Returns:**

`This builder.`

**Throws:**

`IllformedLocaleException - if variant is ill-formed`

### setExtension

```
public Locale.Builder setExtension(char key,
                                   String value)
```

Sets the extension for the given key. If the value is null or the empty string, the extension is removed. Otherwise, the extension must be well-formed or an exception is thrown.

**Note:** The key `UNICODE_LOCALE_EXTENSION` ('u') is used for the Unicode locale extension. Setting a value for this key replaces any existing Unicode locale key/type pairs with those defined in the extension.

**Note:** The key `PRIVATE_USE_EXTENSION` ('x') is used for the private use code. To be well-formed, the value for this key needs only to have subtags of one to eight alphanumeric characters, not two to eight as in the general case.

**Parameters:**

`key - the extension key`

`value - the extension value`

**Returns:**

`This builder.`

**Throws:**

IllformedLocaleException - if key is illegal or value is ill-formed

**See Also:**

setUnicodeLocaleKeyword(String, String)

### setUnicodeLocaleKeyword

```
public Locale.Builder setUnicodeLocaleKeyword(String key,
                                              String type)
```

Sets the Unicode locale keyword type for the given key. If the type is null, the Unicode keyword is removed. Otherwise, the key must be non-null and both key and type must be well-formed or an exception is thrown.

Keys and types are converted to lower case.

**Note**:Setting the 'u' extension via setExtension(char, java.lang.String) replaces all Unicode locale keywords with those defined in the extension.

**Parameters:**

key - the Unicode locale key

type - the Unicode locale type

**Returns:**

This builder.

**Throws:**

IllformedLocaleException - if key or type is ill-formed

NullPointerException - if key is null

**See Also:**

setExtension(char, String)

### addUnicodeLocaleAttribute

```
public Locale.Builder addUnicodeLocaleAttribute(String attribute)
```

Adds a unicode locale attribute, if not already present, otherwise has no effect. The attribute must not be null and must be well-formed or an exception is thrown.

**Parameters:**

attribute - the attribute

**Returns:**

This builder.

**Throws:**

NullPointerException - if attribute is null

IllformedLocaleException - if attribute is ill-formed

**See Also:**

setExtension(char, String)

### removeUnicodeLocaleAttribute

public Locale.Builder removeUnicodeLocaleAttribute(String attribute)

Removes a unicode locale attribute, if present, otherwise has no effect. The attribute must not be null and must be well-formed or an exception is thrown.

Attribute comparision for removal is case-insensitive.

**Parameters:**

attribute - the attribute

**Returns:**

This builder.

**Throws:**

NullPointerException - if attribute is null

IllformedLocaleException - if attribute is ill-formed

**See Also:**

setExtension(char, String)

### clear

public Locale.Builder clear()

Resets the builder to its initial, empty state.

**Returns:**

This builder.

### clearExtensions

public Locale.Builder clearExtensions()

Resets the extensions to their initial, empty state. Language, script, region and variant are unchanged.

**Returns:**

This builder.

**See Also:**

`setExtension(char, String)`

### build

```
public Locale build()
```

Returns an instance of `Locale` created from the fields set on this builder.

This applies the conversions listed in `Locale.forLanguageTag(java.lang.String)` when constructing a Locale. (Grandfathered tags are handled in `setLanguageTag(java.lang.String)`.)

**Returns:**

A Locale.



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